

Muni Koilakuntla

munikoilakuntla@gmail.com | +1 (618) 434-5179 | [linkedin.com/in/muni-koilakuntla-5334851b8](https://www.linkedin.com/in/muni-koilakuntla-5334851b8)

SUMMARY

Full Stack Software Engineer with 5+ years of experience in designing, developing and deploying scalable, cloud-native web applications in financial services, insurance and healthcare domains. Proficient in Java, Python, Spring Boot, Angular, React, SQL, and Azure DevOps, with strong command over both frontend and backend development. Skilled in REST APIs integration, Microservices, database optimization and DevOps automation, with a proven record of delivering high-performance solutions and mentoring teams in Agile environments. Strong grasp of AWS and Azure cloud deployment, CI/CD pipeline, security best practices (OAuth2/JWT), and data-driven problem solving.

SKILLS

- **Programming Languages** : Java, Python, SQL, JavaScript, TypeScript, Bash, Shell Script, HTML5, CSS
- **Frontend Frameworks** : Angular 16, React (TypeScript), Bootstrap, Responsive Web Design
- **Backend Frameworks** : Spring Boot, Spring MVC, Microservices, REST APIs, Hibernate, JPA, JWT, JMS, MQ
- **Databases** : SQL Server, NoSQL, PostgreSQL, MongoDB, Oracle, Relational databases
- **Cloud & DevOps** : AWS (EC2, S3, Lambda), Azure DevOps, GCP, OpenShift, Docker, Jenkins, CI/CD Pipelines
- **Infrastructure & Monitoring** : Terraform, AWS CloudFormation, ELK Stack, Splunk, Prometheus, Grafana
- **MLOps & ML Tools** : Scikit-learn, TensorFlow, Pandas, SageMaker, MLflow, JUnit, SonarQube
- **Tools & Frameworks** : Spring Boot, Visual Studio Code, Guidewire, Gradle, Maven, Git, GitHub, Confluence
- **API & Web Services Testing** : SOAP/REST APIs, JUnit5, Mockito, TestNG, Integration Testing, SonarQube, Postman, Swagger
- **Methodologies** : Agile/ Scrum, SDLC, Test-Driven Development (TDD)

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Jun 2021 - Jan 2024

Capgemini Technology Services, India

- Designed and developed RESTful microservices using Java, Spring Boot, Hibernate, Spring Cloud frameworks, and JPA to deliver scalable, maintainable solutions that met complex business requirements and aligned with enterprise architecture standards.
- Modernized front-end with Angular, TypeScript, and Bootstrap, delivering responsive, modular front-end components that improved client satisfaction by reducing ticket escalations in production environment.
- Integrated MySQL and Hibernate for database design, optimization and management to maintain data integrity and security, ensured ACID compliance and improved query performance.
- Configured Guidewire ClaimCenter modules to automate insurance processes and implemented system integrations using SOAP/RESTful APIs, Web Services and exposed services for internal and external system integrations.
- Built and deployed containerized applications using Docker, Kubernetes, and Jenkins, enabling zero-downtime deployments and accelerating release cycles and reducing deployment failures.
- Developed and optimized event-driven and message-driven architectures using Apache Kafka and RabbitMQ, enabling real-time communication between distributed services.
- Provided L3 production support by troubleshooting issues, analyzing logs, and performing root cause analysis; applied code reviews, refactoring, and design patterns to resolve 40+ high-priority production issues.
- Developed messaging-driven integrations using JMS and MQ to enable asynchronous communication across core services, enhancing system reliability and scalability.
- Created Unit test cases using JUnit and Mockito, collaborated with QA teams for comprehensive testing across UAT and integration test environments, enabling 3x faster release cycles and ensuring robust performance, scalability, and reliability.
- Partnered with DevOps teams to build and manage CI/CD pipelines in Jenkins and Azure DevOps, integrating automated testing, code quality checks (SonarQube), and deployment automation.
- Collaborated with cross-functional teams including Business Analysts, product managers and stakeholders in Agile Sprint sessions to translate business requirements into technical designs, contributing to sprint planning, daily scrum meetings to deliver high quality software solutions.
- Prepared technical documentation, presentations and architecture deliverables, facilitating stakeholder understanding and alignment in Agile/Scrum environments.
- Mentored junior engineers and led knowledge-sharing sessions on microservices best practices, DevOps automation, and API integration patterns.
- Well-versed in SDLC, Agile methodology, Version control system (GIT), Defect Tracking tool (Jira).

Graduate Teaching Assistant
Southern Illinois University Carbondale, IL

- Mentored 60+ students in Agile-driven cloud-native deployments leveraging Java, Spring Boot, React, SQL, REST APIs integration, Object-oriented programming concepts, Microservices and workflows.
- Conducted training sessions simulating software development lifecycle (SDLC) methodologies, guiding teams in API design, data modelling, and CI/CD pipelines similar to real-world enterprise-grade applications.
- Conducted knowledge transfer sessions on RESTful API design, microservices, Kafka event-driven architecture, and GraphQL integrations, API Gateway best practices, preparing students for secure, scalable, and observable distributed systems.
- Contributed to full stack academic industry aligned projects by guiding designing, developing, and deployment efforts on AWS, Azure DevOps, Docker, and Kubernetes, ensuring smooth delivery of projects.

Jan 2024 - Present

Software Engineer - Development Intern
WTA Studios Private Limited, India

- Contributed to the development of microservice-based backend modules using Java and Spring Boot, integrating REST APIs and JWT authentication for secure communication between client and server components.
- Implemented CI/CD automation pipelines with Jenkins and Docker, and explored modern frontend frameworks (Angular) to deliver full stack prototypes aligned with enterprise DevSecOps and cloud-native practices.
- Coordinated with senior developers to optimize SQL queries, triggers, and stored procedures, reducing data retrieval time and supporting faster reporting in internal tools.
- Assisted QA efforts by writing unit test cases and basic automation scripts achieving 75% coverage, contributed to build stability and reduced manual rework through test-driven development (TDD) practices in integration cycles.

Jun 2019 – January 2020

EDUCATION

• Master of Science in Computer Science Southern Illinois University Carbondale, IL	Expected December 2025 GPA 3.9/4.0
• Bachelor of Science in Computer Science Vel Tech University, Chennai, India	June 2017 - May 2021 CGPA 9.2/10

ACADEMIC PROJECTS

Employee Payroll Management System

- Engineered a full stack web-based payroll application using Python, Django, and React, PostgreSQL, and GraphQL APIs to enable efficient role-based access, employee data handling, and payroll automation.
- Integrated user authentication using JWT and exposed REST endpoints, leveraging PostgreSQL for structured data storage and Apache Kafka to process payroll events asynchronously, improving system scalability and resilience.
- Designed and trained a lightweight ML model (Python scikit-learn) for anomaly detection in payroll data such as duplicate entries, unusual deductions, reducing manual review time by 30%.

Mar 2025 – May 2025

Online Eye Clinic System

- Built an enterprise-grade clinical management system application using Java, Spring Boot, and Angular, and MySQL supporting appointment scheduling, patient onboarding, and medical record digitization.
- Designed and validated secure API contracts using Postman and Swagger, enabling cross-layer communication between client-side and server-side components with low-latency inter-service communication.
- Developed custom service logic with JPA and SQL, enabling dynamic data access patterns for doctors, patients, and treatment records and implemented Kafka-based event-driven workflows to manage patient notifications.
- Conducted backend verification using JUnit, ensuring code robustness and achieving 90% test coverage across controller and service layers.

Feb 2021 - May 2021